**Python Activity Record: Retail Billing System**

Based on your uploaded retail building.py program, here is a simple, first-year B.Tech-level activity record. The program calculates product amounts, applies a 10% discount when the subtotal is ₹1,000 or more, adds 5% tax after the discount, and displays the final invoice.

retail building.py

**1. Problem statement**

To develop a Python program that generates a retail invoice by accepting product names, prices, and quantities. The program calculates the subtotal, applies a discount when applicable, adds tax, and displays the final amount payable by the customer.

**2. Objectives**

- To accept product details from the user.
- To calculate the amount for each product.
- To calculate the subtotal of all products.
- To apply a 10% discount if the subtotal is ₹1,000 or more.
- To calculate 5% tax after applying the discount.
- To display the invoice and final amount.

**3. Assumptions**

1. The user enters a valid number of products.
2. Product prices are entered as numbers and quantities as integers.
3. Prices and quantities are non-negative.
4. A 10% discount is applied when the subtotal is at least ₹1,000.
5. Tax is calculated at 5% on the subtotal after discount.
6. The program does not include additional charges or other taxes.

Note: These are assumptions for this program; the code does not validate every input.

**4. Algorithm**

1. Start the program.
2. Create empty lists for product names and amounts.
3. Read the number of products.
4. Repeat for each product:
    - Read the product name.
    - Read its price and quantity.
    - Calculate price × quantity.
    - Store the name and amount.
5. Calculate the subtotal.

6. If the subtotal is ₹1,000 or more, calculate a 10% discount; otherwise, set the discount to zero.
7. Calculate 5% tax on the discounted subtotal.
8. Calculate the final amount by subtracting the discount and adding tax.
9. Display each product amount, subtotal, discount, tax, and final amount.
10. Stop the program.

**5. Flowchart**

Start

Input number of products

Enter product name, price and quantity

Calculate each product amount and subtotal

Is subtotal ≥ ₹1,000?

Yes

Calculate 10% discount

No

Discount = ₹0

Calculate 5% tax after discount

Calculate and display final invoice

Stop

**6. Pseudocode**

```
START

Create empty lists: names and amounts

INPUT number of products

FOR each product:
    INPUT product name
    INPUT price
    INPUT quantity
    amount = price * quantity
    Store product name and amount
END FOR
```

```
subtotal = sum of all amounts

IF subtotal >= 1000:
    discount = subtotal * 0.10
ELSE:
    discount = 0
END IF

tax = (subtotal - discount) * 0.05
total = subtotal - discount + tax

DISPLAY product names and amounts
DISPLAY subtotal
DISPLAY discount
DISPLAY tax
DISPLAY final amount

STOP
```

**7. Python source code**

```
names = []
amounts = []

n = int(input("How many products? "))

for i in range(n):
    name = input("Product name: ")
    price = float(input("Price: "))
    qty = int(input("Quantity: "))

    names.append(name)
```

```
    amounts.append(price * qty)

subtotal = sum(amounts)

if subtotal >= 1000:
    discount = subtotal * 0.10
else:
    discount = 0

tax = (subtotal - discount) * 0.05
total = subtotal - discount + tax

print("\\n--- INVOICE ---")

for i in range(n):
    print(names[i], ": ₹", round(amounts[i], 2))

print("Subtotal: ₹", round(subtotal, 2))
print("Discount: ₹", round(discount, 2))
print("Tax: ₹", round(tax, 2))
print("Final amount: ₹", round(total, 2))
```

**8. Test cases**

| Test | Product details | Subtotal | Discount | Tax | Final amount |
|---|---|---|---|---|---|
| 1 | Pen: ₹100 × 2; Bag: ₹500 × 2 | ₹1,200 | ₹120 | ₹54 | ₹1,134 |
| 2 | Book: ₹200 × 2; Pen: ₹100 × 4 | ₹800 | ₹0 | ₹40 | ₹840 |
| 3 | Shoes: ₹500 × 2 | ₹1,000 | ₹100 | ₹45 | ₹945 |

All three test cases use valid inputs and follow the discount and tax rules in the program.

**9. Output screenshots**

The following are sample console output previews, not actual screenshots. You can run the program in Python IDLE or VS Code and capture the real output using a screenshot.

**Test case 1 — Discount applied**

Sample input: 2 products, Pen ₹100 × 2, Bag ₹500 × 2

How many products? 2
Product name: Pen
Price: 100
Quantity: 2
Product name: Bag
Price: 500
Quantity: 2

--- INVOICE ---
Pen : ₹ 200.0
Bag : ₹ 1000.0
Subtotal: ₹ 1200.0
Discount: ₹ 120.0
Tax: ₹ 54.0
Final amount: ₹ 1134.0

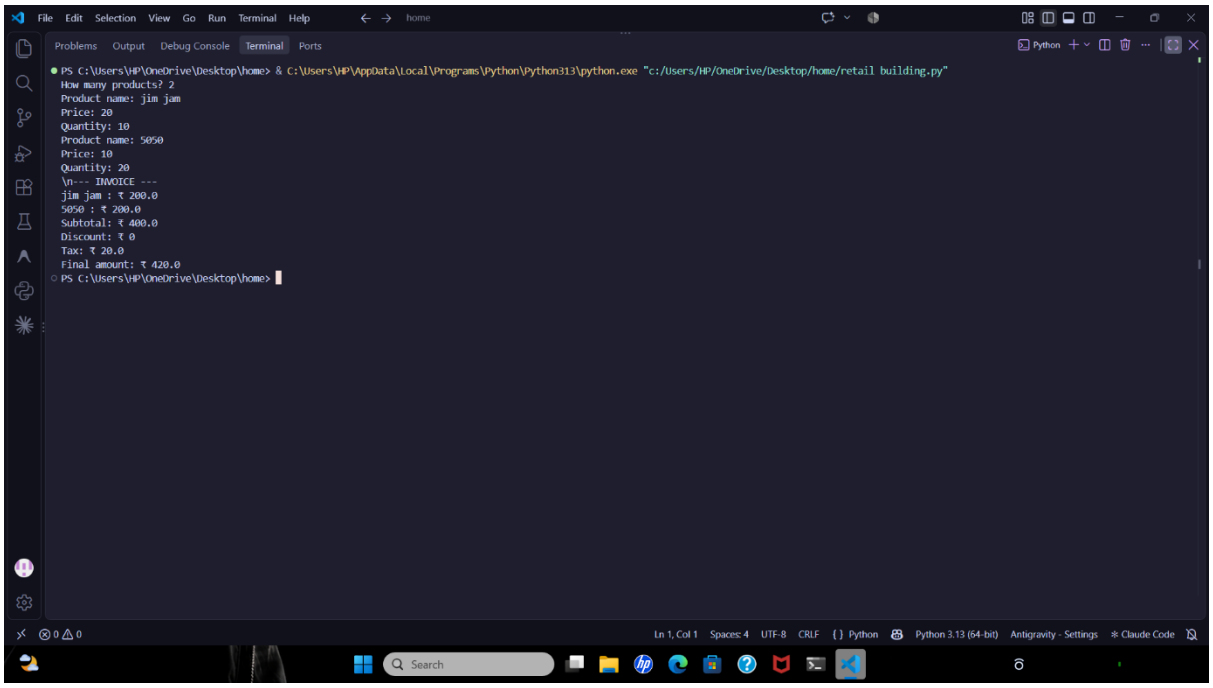
Copy sample output

**Test case 2 — No discount**

Sample input: 2 products, Book ₹200 × 2, Pen ₹100 × 4

--- INVOICE ---
Book : ₹ 400.0
Pen : ₹ 400.0
Subtotal: ₹ 800.0
Discount: ₹ 0
Tax: ₹ 40.0
Final amount: ₹ 840.0

Copy sample output




**10. Conclusion**

The Retail Billing System was developed successfully using Python. It accepts product details, calculates individual product amounts and the subtotal, applies a discount when the subtotal reaches ₹1,000, and adds 5% tax after the discount. Finally, it displays the invoice and total amount payable. This program demonstrates the use of lists, loops, conditional statements, arithmetic operations, and built-in functions in Python.